LÍNGUA INGLESA

T E X T

India launches space mission to the sun a week after moon landing

The latest mission in India's ambitious space program has blasted off on a voyage towards the center of the solar system, a week after the country's successful unmanned moon landing.

Aditya-L1 launched shortly before midday, with a live broadcast showing hundreds of spectators cheering wildly against the deafening noise of the rocket's ascent.

"Launch successful, all normal," an Indian Space Research Organization official announced from mission control as the vessel made its way to the upper reaches of the Earth's atmosphere.

The mission is carrying scientific instruments to observe the sun's outermost layers in a four-month journey.

The US and the European Space Agency (Esa) have sent numerous probes to the centre of the solar system, beginning with Nasa's Pioneer program in the 1960s.

Japan and China have both launched their own solar observatory missions into Earth's orbit.

But if successful, the latest mission from the Indian Space Research Organization (ISRO) will be the first by any Asian nation to be placed in orbit around the sun.

"It's a challenging mission for India," astrophysicist Somak Raychaudhury told broadcaster NDTV on Friday.

Raychaudhury said the mission probe would study coronal mass ejections, a periodic phenomenon that sees huge discharges of plasma and magnetic energy from the sun's atmosphere.

These bursts are so powerful they can reach the Earth and potentially disrupt the operations of satellites.

Aditya would help predict the phenomenon "and alert everybody so that satellites can shut down their power", he said.

"It will also help us understand how these things happen, and in the future, we might not need a warning system out there."

Aditya, the name of the Hindu Sun deity, will travel 1.5m km (930,000 miles) to reach its destination – still only 1% of the vast distance between Earth and the sun.

At that point, the gravitational forces of both celestial bodies cancel each other out, allowing the mission to remain in a stable halo orbit around our nearest star.

Aditya is travelling on the ISRO-designed, 320-tonne PSLV XL rocket that has been a mainstay of the Indian space program, powering earlier launches to the moon and Mars.

The mission also aims to shed light on the dynamics of several other solar phenomena by imaging and measuring particles in the sun's upper atmosphere.

India has been steadily matching the achievements of established spacefaring powers at a fraction of their cost.

The South Asian nation has a comparatively low-budget space program, but one that has grown considerably in size and momentum since it first sent a probe to orbit the moon in 2008.

Experts say India can keep costs low by copying and adapting existing technology, and thanks to an abundance of highly skilled engineers who earn a fraction of their foreign counterparts' wages.

Last month's successful landing on the lunar surface – a feat previously achieved only by Russia, the US and China – cost less than $75m.

The touchdown was widely celebrated by the public, with prayer rituals to wish for the mission's success and schoolchildren after its final descent from live broadcasts in classrooms.

India became the first Asian nation to put a craft into orbit around Mars in 2014 and is slated to launch a three-day crewed mission into the Earth's orbit by next year.

It also plans a joint mission with Japan to send another probe to the moon by 2025 and an orbital mission to Venus within the next two years.

Adapted from: www.theguardian.com/world/2023/sep/02/

01. On its four-month journey to the sun, the Aditya-L1 mission will study
A) solar flares.
B) coronagraphic imaging.
C) coronal mass ejections.
D) solar eclipses.

02. The specific purpose of Aditya-L1's study of coronal mass ejections is to
A) study the sun's magnetic fields for space travel advancements.
B) improve satellite communication and prevent signal disruptions.
C) understand the sun's core processes and their impact on Earth.

D) identify potential fuel sources for future space missions.

03. India's space program manages to reduce costs in comparison to other nations by
A) hiring foreign engineers.
B) using expensive technology.
C) adapting existing technology.
D) maintaining a large budget.

04. In terms of its mission specifics and cost, India's lunar landing is distinguished by the fact that it
A) was accomplished at a lower cost.
B) was the first to discover lunar water.
C) marked the first manned lunar landing.
D) successfully brought lunar samples back to Earth.

05. As mentioned in the text, the specific role of the PSLV XL rocket in India's space program is to
A) transport satellites to Earth's orbit.
B) power lunar landings.
C) send missions to Mars.
D) launch solar observatory missions.

06. The significance of naming the mission "Aditya" is because it
A) is named after the project lead.
B) signifies the mission's destination.
C) reflects the mission's cost-saving approach.
D) is derived from a Hindu deity.

07. The term "coronal mass ejections" refers to
A) massive solar explosions that can affect satellite operations.
B) lunar eclipses and has no impact on Earth.
C) a type of solar eclipse visible from space.
D) a type of solar wind that helps spacecraft propulsion.

08. In collaboration with Japan, India is planning
a) a mission to study Earth's magnetic field.
b) a crewed mission to Mars.
c) an orbital mission to the sun.
d) another mission to the moon.

VOCABULARY

Adapting - Adaptando
Ascent - Ascensão
Astrophysicist - Astrofísico
Atmosphere - Atmosfera
Burst - Explosão
Celestial bodies - Corpos celestes
Center - Centro
Challenging - Desafiador
Cheering - Torcendo
Control - Controle
Copying - Copiando
Coronagraphic imaging - Imagem coronográfica
Coronal mass ejections - Ejeções de massa coronal
Cost less - Custo menor
Cost-saving - Economia de custos
Cost-saving approach - Abordagem de economia de custos
Costs - Custos
Crewed mission - Missão tripulada
Designed - Projetado
Destination - Destino
Discharges - Descargas
Disrupt - Interromper
Earn - Ganham
Earth's atmosphere - Atmosfera da Terra
Earth's orbit - Órbita da Terra
Engineers - Engenheiros
European Space Agency - Agência Espacial Europeia
Existing technology - Tecnologia existente
Final descent - Descida final
First - Primeiro
Four-month journey - Jornada de quatro meses
Fraction - Fração
Gravitational forces - Forças gravitacionais
Halo orbit - Órbita de halo
Highly skilled engineers - Engenheiros altamente qualificados
Hindu Sun deity - Deidade hindu do Sol
Imaging - Imagens
Joint mission - Missão conjunta
Kilometers - Quilômetros
Landing - Pouso
Launch - Lançamento
Live broadcast - Transmissão ao vivo
Live broadcasts - Transmissões ao vivo
Low-budget - Orçamento baixo
Lunar landing - Pouso lunar
Lunar probe - Sonda lunar
Lunar samples - Amostras lunares
Magnetic energy - Energia magnética
Mainstay - Esteio
Manned mission - Missão tripulada
Mars - Marte
Measuring - Medição
Miles - Milhas
Mission - Missão
Moon - Lua
Nearest star - Estrela mais próxima

Observe - Observar
Operations - Operações
Orbit - Órbita
Orbital mission - Missão orbital
Outermost layers - Camadas mais externas
Particles - Partículas
Phenomenon - Fenômeno
Placed in orbit - Colocado em órbita
Plasma - Plasma
Powerful - Poderosas
Prayer rituals - Rituais de oração
Predict - Prever
Probes - Sondas
Public - Público
Rocket - Foguete
Rocket - Foguete
Satellite communication - Comunicação por satélite
Satellites - Satélites
Schoolchildren - Estudantes
Scientific instruments - Instrumentos científicos
Slated to launch - Programado para lançamento
Solar flares - Ejeções solares
Solar observatory missions - Missões de observação solar
Solar phenomena - Fenômenos solares
Solar system - Sistema solar
Space - Espaço
Space travel advancements - Avanços em viagens espaciais
Spectators - Espectadores
Stable - Estável
Successful - Bem-sucedido
Sun - Sol
Travel - Viajar
Unmanned - Não tripulado
Venus - Vênus
Vessel - Nave
Warning system - Sistema de alerta
Wish - Desejo